Milestone 0: Timeline & Project Planning

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Table of Contents

- Timeline
  - Stakeholder Milestones
  - Course Deadlines
- Milestone Planning
  - Milestone 0: Timeline & Project Planning
  - Milestone 1: Preliminary Research & Design Inputs
  - Milestone 2: Research & Design Planning
  - Milestone 3: Component Prototyping & Design Outputs
  - Milestone 4: Final Implementation Plan
- Appendix A: Progress Report Instructions
  - Progress Report 1
    - Background and Motivation
    - User needs and project statement
  - Progress Report 2
    - Design Inputs
    - Development Process & Preliminary Plan
    - Preliminary Risk Analysis
  - Progress Report 3
    - Summary
    - Design Outputs
    - Final concept and preliminary design configuration
    - Risk and compliance management plan

Timeline

Stakeholder Milestones

Week #	Dates	Milestone Tasks	Project Manager
1	2026-09-26 - 2026-10-02	M0: Timeline & Project Planning	Seaya
2-4	2026-10-03 - 2026-10-23	M1: Preliminary Research & Design Inputs	Ursula (co-lead with Kyle)
5-6	2026-10-24 - 2026-11-06	M2: Research & Design Planning	Kyle (co-lead with Ursula)
7-9	2026-11-07 - 2026-11-20	M3: Component Prototyping & Design Outputs	Baoze

9-10	2026-11-21 - 2026-12-04	M4: Final Implementation Plan	Seaya

Course Deadlines

Week #	Deadline	Deliverable	Complete With
1	2026-10-02	Progress Report 1	M0
2-4	2026-10-23	Progress Report 2	M1
5-6	2026-10-29	Graded Advisor Meeting 1	M2
9-10	2026-11-26	Graded Advisor Meeting 2	M3
9-10	2026-12-03	Proof of Concept (POC) Demo Deadline	M3 + M4
>10	2026-12-11	Progress Report 3	M4

Milestone Planning

Milestone 0: Timeline & Project Planning

Due: 2026-10-02

Lead: Seaya

Milestone Deliverables

Create a fall semester timeline for stakeholder milestones, defining research topics, user needs, rough deadlines, and project manager assignments.

Course Deliverables

1. **Progress Report 1** is due on **October 2nd**, focusing on the topics:
    - Background and motivation (Ursula)
    - User needs and project statement (Seaya)
    - Requirements are listed at Appendix A — Progress Report 1

Milestone 1: Preliminary Research & Design Inputs

Due: 2026-10-23

Lead: Ursula

Complete preliminary research on the following topics to guide design input decisions and development planning:

1. Doppler ultrasound (B-mode, Doppler, FMD)
    - **Assignee:** Baoze
    - Determine what we need to test (which kinds of ultrasounds - ex. interventions vs standard, how many and type of test subjects)
    - Bonus: ECG (Pan-Tompkins Algorithm)
2. Danny Green's software

    - **Assignee:** Ursula
    - Written in 4tran, can do perpendicular diameter measurement, cannot do cross sectional area
    - Does not meet all user need requirements
3. PhysioMerge integration
    - **Assignee:** Kyle
    - What functions/commands can we use? How will we use them?
    - What additional features does it enable us to add to our software?
4. Image processing tools & libraries
    - **Assignee:** Seaya
    - Research commonly used tools for edge detection
    - Are there specific tools better suited for ultrasound imaging?

Milestone Deliverables

Requirements & data

- List image processing requirements and robustness cases (Seaya)
- Determine hardware constraints (GPU/CPU, real-time) (Baoze)
- Design a test dataset with ground truth (Baoze, Ursula, Kyle)
- Write a validation test plan against Danny Green's software (Ursula)
- Brainstorm video segmentation methods (Seaya / All)
- List requirements for test harness (All)

GUI & PhysioMerge

- List UI requirements (All)
- PhysioMerge recon: data handling, available commands (Kyle, `Baoze)
- Write a PhysioMerge integration spec (Kyle, Baoze)
- List specific requirements for output (All)

Course Deliverables

1. **Progress Report 2** is due on **October 23rd**, focusing on the topics:
    - Updated user and problem framing (Seaya)
    - Design inputs (Baoze)
    - Concept-development process and preliminary analysis (All)
    - Preliminary risk analysis (Ursula)
        - Requirements are listed at Appendix A — Progress Report 2
2. A **graded advisor meeting** is scheduled for **October 29th**, with Dr. McDonald, Dr. Smith, and Mohamad. **The meeting agenda must be submitted to the advisors *one week* before the scheduled meeting, on October 22nd.** See Milestone 2 course deliverables for more details on meeting requirements.

Milestone 2: Research & Design Planning

Due: 2026-11-06

Lead: Kyle

Complete and compile research from Milestone 1 towards defining and detailing specific project tickets. Refine and iterate on the design plan based on continual stakeholder feedback, determine design outputs, create a rough stage-based timeline, and discuss responsibility divisions. Determine if any materials or resources need to be requested for the prototyping stage (and future development stages).

Milestone Deliverables

Segmentation & test harness

- Compare segmentation methods (Seaya)
- Throwaway test: segment fixed regions on the test dataset (Seaya)
- Design the test harness (Ursula, Seaya)

Throwaway tests for high-risk algorithms

- Doppler: OCR the scale (Kyle)
- Doppler: extract the waveform and its envelope (Ursula)
- Doppler: separate the ECG and detect R-peaks (Ursula)
- Doppler: track the timeline from timestamps (Baoze)
- B-mode: OCR the depth scale (Kyle)
- Longitudinal: detect artery walls (Seaya)
- Longitudinal: calculate angle-corrected diameter (Seaya)
- Transverse: detect circular vessels (Baoze)

UI

- Define UI states (process, edit, pause, cancel, undo) (Baoze)
- Wireframe the UI and start the Figma overlay (Baoze, Seaya)

Course Deliverables

1. A **graded advisor meeting** is scheduled for **October 29th**, with Dr. McDonald, Dr. Smith, and Mohammed.

   The graded instructor meetings evaluate the quality of the team’s engineering design process across the full year. They provide recurring checkpoints for students to demonstrate preparation, technical progress, engineering judgement, responsiveness to evidence and feedback, and a credible plan for subsequent work.

   At this stage of the project, the team is expected to have preliminary ideas and research completed to present. The meeting will also be used for eliciting feedback and expertise from the instructors.

Milestone 3: Component Prototyping & Design Outputs

Due: 2026-11-20

Lead: Baoze

Milestone Deliverables

Foundation

- Prototype automatic region detection (Seaya)
- Implement and verify the test harness (Seaya, Ursula)
- Define the output interface, including how to handle undesirable data (Baoze)

Doppler Throwaway Tests

- Extract metrics: velocity, systole/diastole, pulsatility index (Ursula, Baoze)

Longitudinal View Throwaway Tests

- Isolate the green placement box (Seaya, Baoze)
- Track diameter across frames (Seaya)

Transverse view

- Prototype vessel detection and CSA (Kyle, Baoze)
- Track CSA across frames (Baoze)

GUI

- Finish the Figma overlay (Baoze)
- Prototype the overlay on the FMD video (Baoze)

All prototypes should work by the end of this milestone.

Course Deliverables

1. A **graded advisor meeting** is scheduled for **November 26th**, with Dr. McDonald, Dr. Smith, and Mohamad. **The meeting agenda must be submitted to the advisors one week before the scheduled meeting, on November 19th.**
2. The **POC demo** will take place **between December 3rd and December 10th**. For our timeline purposes we are aiming to complete corresponding tasks and deliverables for December 3rd. At this time (2026-10-02), the demo rubric has not been released, so the following tasks are high-level ideas aligned with what we would want to test/prototype in these planning stages for our end product:
    - Demo image processing library/tool tests with initial FMD imaging (and other data if possible) → discuss the results of each tool and which ones will be used in future development of the product.
    - Present UI prototyping (ex. Figma) and vision for PhysioMerge integration.

Milestone 4: Final Implementation Plan

Due: 2026-12-04

Lead: Seaya

Compile research, prototypes, decisions, and design plan drafts into one comprehensive final implementation plan. This plan should meet all the requirements outlined in Final concept and preliminary design configuration in Appendix A.

Milestone Deliverables

- Finalize Doppler Design (Ursula, Kyle)
- Finalize B-Mode Design (Seaya, Kyle, Baoze)

- Finalize UI (Baoze)
- Finalize the PhysioMerge integration design (All)
- Compile the final design document (All)
- Compile the final presentation/demo (All)
- Evaluate feasability of project extensions (All)

Course Deliverables

1. A **graded advisor meeting** is scheduled for **November 26th**, with Dr. McDonald, Dr. Smith, and Mohamad.

   At this stage of the project, the team is expected to present a preliminary prototype and explain the design outputs that are chosen. Instructors use this meeting to provide constructive criticism and challenge the team to consider scenarios that have yet to be implemented. The team will consolidate and refine the prototype before the POC demonstration.

2. **Progress Report 3** is due on **December 11th**, focusing on topics: (responsibility divisions are TBD)

   - Summary (background, user inputs and solution)
   - Design outputs
   - Final concept and preliminary design configuration
   - Risk and compliance management plan
   - Requirements are listed at Appendix A — Progress Report 3

3. **Proof of Concept Demo** occurs on **December 3rd**, focusing on the following (All):

   - Explanation of design choices and assumption
   - Showcasing prototype of project
   - Feedback from professors and teaching assistants

Appendix A: Progress Report Instructions

Section requirements copied from the IBEHS 5P06 progress report instructions. Each milestone above links to the sections it covers.

Progress Report 1

Due: 2026-10-02 — see Milestone 0

Background and Motivation

- Describe the problem context and the circumstances that motivate the project.
- Use stakeholder discussions and relevant literature to explain why the problem is important and why an engineering response is warranted.
- Define the intended user(s), stakeholder(s), and setting of use with enough specificity to guide later design decisions.

User needs and project statement

- Summarize the user needs that your team has identified and distinguish needs from proposed solutions.
- Explain the expected impact of a successful design on the stakeholder, broader society, or environment. Consider health, safety, economic, social, environmental, legal, and cultural

dimensions where relevant.
- End with a short, explicit need statement that is visually distinct from the surrounding text (e.g. bolded, italicized, on a separate line etc.)

Progress Report 2

Due: 2026-10-23 — see Milestone 1

Design Inputs

- Define the requirements, specifications, objectives, and constraints that the design is expected to satisfy.
- Develop objective and, wherever feasible, quantitative criteria through stakeholder consultation, literature review, and discussions with disciplinary instructors.
- Consider technical, economic, environmental, ergonomic, societal, safety, legal, and regulatory inputs as applicable.
- Trace each important design input to a user need, stakeholder expectation, risk, standard, or other credible source

Development Process & Preliminary Plan

- Describe the structured process your team will use to generate, analyze, and compare multiple design concepts.
- Use relevant theory, calculations, models, or simulations to investigate feasibility to an appropriate preliminary depth.
- Consider uncertainty and trade-offs, including societal, environmental, and legal factors that could affect concept selection.

Preliminary Risk Analysis

- Identify early health and safety risks associated with implementation, prototyping, verification, manufacturing, deployment, or use.
- Identify potentially applicable standards, codes, laws, and regulatory requirements, including the consequences of non-compliance.
- Identify relevant ethical concerns and their possible effects on users or society.
- Comment on how the identified risks or constraints influence design concept generation and evaluation

Progress Report 3

Due: 2026-12-11 — see Milestone 4

Summary

- Provide a concise overview of the problem, user need, selected solution direction, planned implementation, and expected impact.
- Include only the background from earlier reports that is necessary to understand this stage of the project.

Design Outputs

- Present detailed, objective, and appropriately quantitative design outputs derived from the design inputs.
- Justify outputs using stakeholder evidence, literature, engineering analysis, standards, and disciplinary guidance.
- Address technical, economic, environmental, ergonomic, societal, safety, and regulatory considerations where relevant.

Final concept and preliminary design configuration

- Justify selection of the final design concept by tracing the decision to user needs, design inputs, evaluation criteria, analytical work, and documented trade-offs.
- Show a credible starting configuration using calculations, models, simulations, low-fidelity prototypes, schematics, pseudocode, or working demonstrations as appropriate.
- Incorporate the findings of your proof-of-concept to demonstrate the feasibility of your final design concept.
- Explain assumptions, limitations, constraints, feasibility, required resources, unanswered questions, and next steps.
- If an output will not be implemented, identify it explicitly and justify the omission using documented evidence.
- If the user need or project framing changed, explain the evidence and obtain instructor approval for the change.

Risk and compliance management plan

- Meet with your disciplinary instructor to identify and approve an appropriate risk management tool for your project. Use the approved tool to develop a risk management plan. If no specific tool is recommended, a Failure Mode and Effects Analysis (FMEA) is used.
- Identify health and safety risks associated with implementation, prototyping, verification, manufacturing, deployment, and use.
- Define feasible risk-mitigation and compliance strategies tied to applicable standards, codes, laws, and regulatory requirements.
- State which strategies must be implemented and verified in the final prototype.